SPOT PLUS DASHBOARD

Product Presentation – Rev. 3.0

October 17, 2025

Advanced Automotive Solutions

DESIGN PROPOSAL







Pictures are just an example and should not be considered definitive

DESIGN PROPOSAL: FRONT VIEW

5 telltales

3.5’’ color TFT
with bonded cover glass



Telltales solution:

- White LEDs on the PCBA
- Dial plate with colors and symbols of telltales
- Transparent cover glass
- A univocal electronic hardware population to satisfy different product configurations only changing the dial plate

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DESIGN PROPOSAL: REAR VIEW

Gore valve and buzzer housing

26-pin connector, Tyco Super Seal compatible

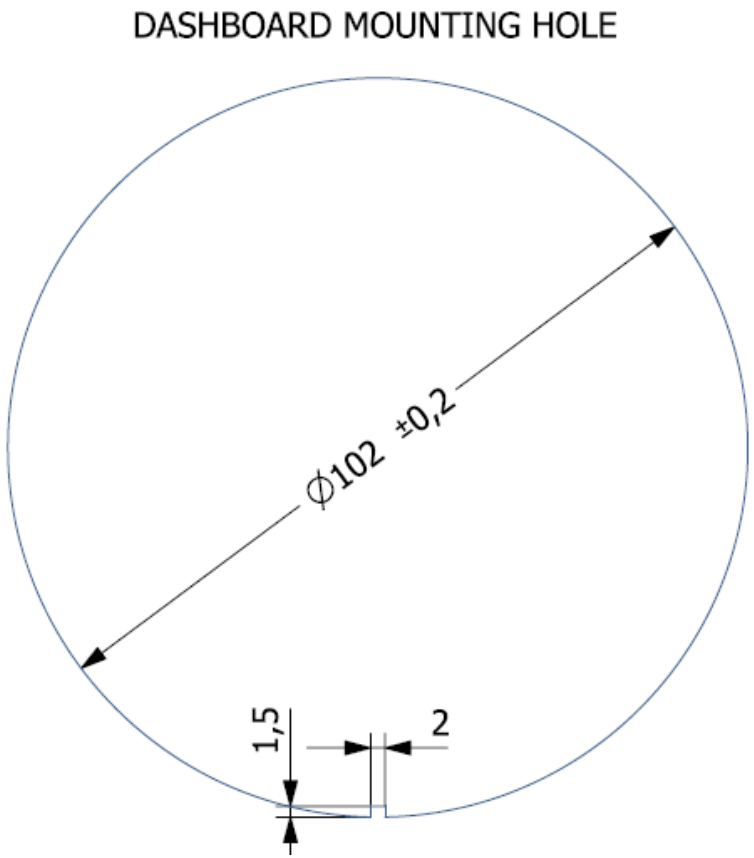


Label

Connector:
The mating connector used for wiring harness is a Tyco 26 pins.
The following parts are required to assemble a wiring harness
connector: 26 pins, 3 keys, connector
PN#: 3-1437290-7

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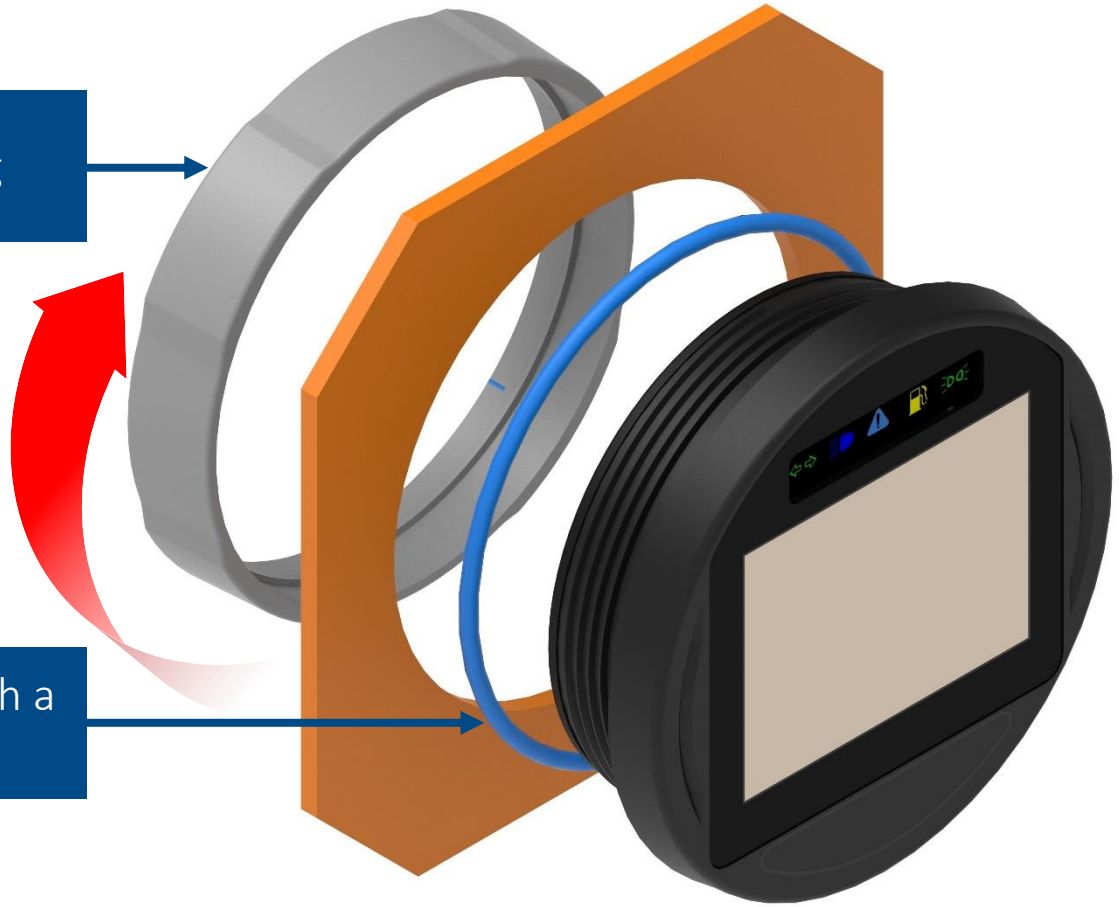
DESIGN PROPOSAL: INSTALLATION



Recommended lockable thickness between 1 – 7 mm

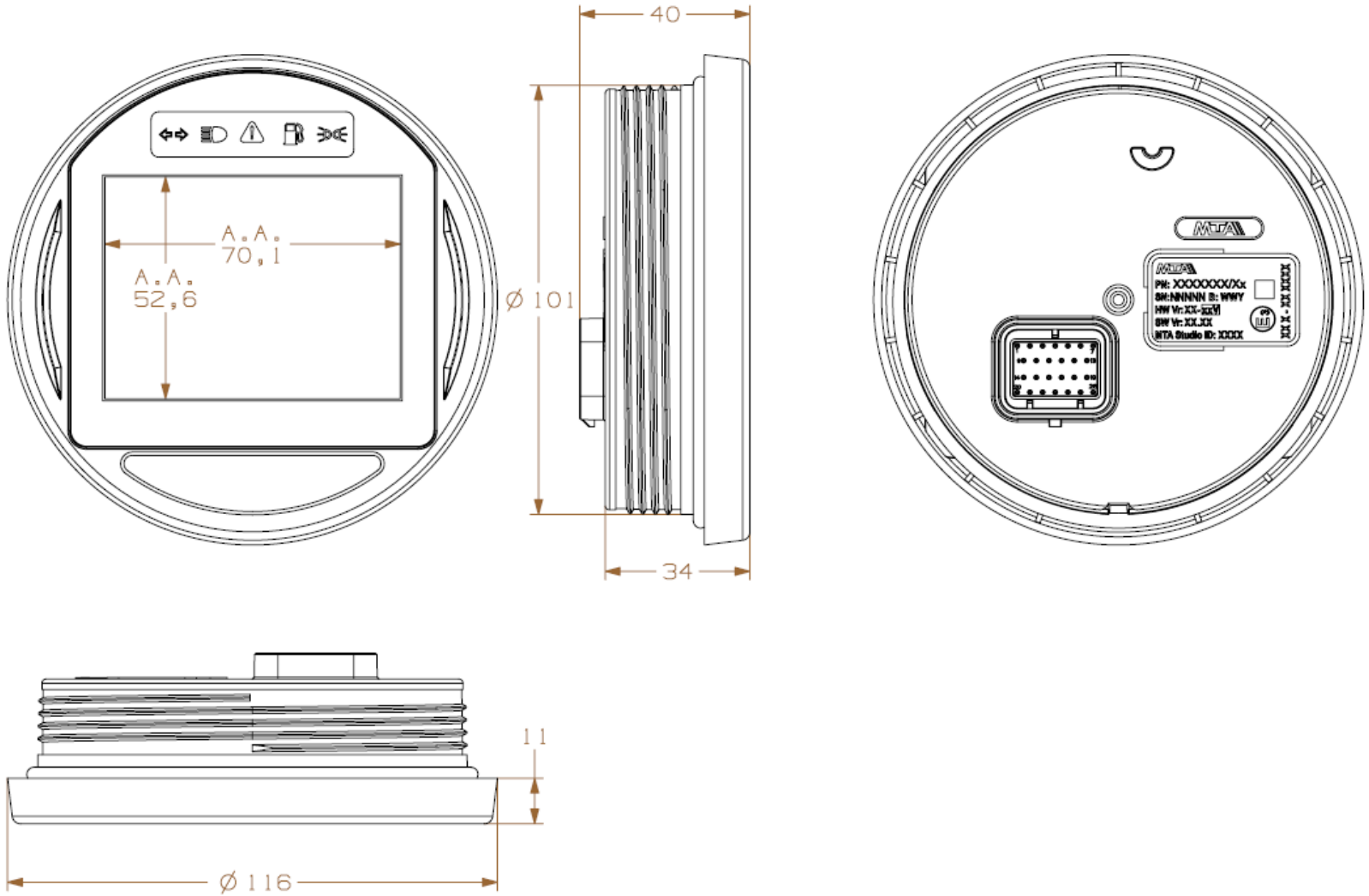
Dashboard fixing ring

Dashboard coupling with a rubber gasket



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DESIGN PROPOSAL: DIMENSIONS



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PINOUT

The pinout is the same as the SPOT already in production.
Only the software configurability of some pins changes.

PIN	NAME	TYPE
1	D+ Alternator	Alternator input
2	IN_17	Analog input with pull-up
3	IN_16	Analog input with/without pull-up*
4	IN_6	Digital input low side/high side*
5	IN_5	Digital input low side/high side*
6	IN_4	Digital input low side/high side*
7	IN_3	Digital input low side/high side*
8	IN_2	Digital input low side/high side*
9	IN_1	Digital input low side/high side*
10	IN_15	Digital input low side
11	IN_14	Digital input low side
12	IN_13	Digital input low side
13	IN_12	Digital input low side
14	IN_11	Digital input low side
15	IN_19	Frequency input with pull-up/pull-down*
16	IN_10	Digital input high side with wake-up
17	IN_9	Digital input high side with wake-up
18	IN_8	Digital input high side direct to telltale L5
19	IN_7	Digital input high side direct to telltale L1
20	IN_18	Frequency input with pull-up
21	CAN_L	CAN bus low
22	CAN_H	CAN bus high
23	GND (terminal 31)	Battery ground
24	OUT_0	Low side output 0.5A max
25	VB (terminal 30)	Battery power supply
26	VCTRL (terminal 15)	Ignition input

NOTE: Pins marked with * can bo configurable via software

MAIN CHARACTERISTICS

MICROCONTROLLER
- Infineon Traveo II CYT3DL Automotive grade
- 240 MHz 32-bit Arm Cortex M7 CPU and 100 MHz 32-bit Arm Cortex –M0+ CPU
- Memory: 4,16 MB Flash + 512 KB RAM
- Video RAM: 2 MB

EXTERNAL MEMORIES
- 32 MB Flash

COMMUNICATION
- 1 CAN bus line

MEMORY
- EEPROM 32 Kbyte

OPTICAL
- 5 telltales with white LEDs

BUZZER
- Internal Buzzer, 80 dB @ 30 cm

RTC (Real time clock)
- Estimated time with RTC keeping without battery: 10 days @25 °C typical

WAKE-UP
- From key, CAN line and from 2 digital inputs

POWER SUPPLY
- 12/24 Vdc extended range power supply

MAIN CHARACTERISTICS

TEMPERATURE RANGE

- Operational: - 30 ÷ + 75 °C
- Storage: - 40 ÷ + 85 °C

PROTECTION DEGREE

- IP69K front
- IP65 rear (guaranteed with harness connector mated with same or higher IP grade)

INPUTS

- 6x digital high side/low side inputs software configurable
- 5x digital inputs low side
- 4x digital inputs high side
- 2x analog inputs
- 2x frequency inputs
- 1x D+ (alternator) analog input with excitation resistor

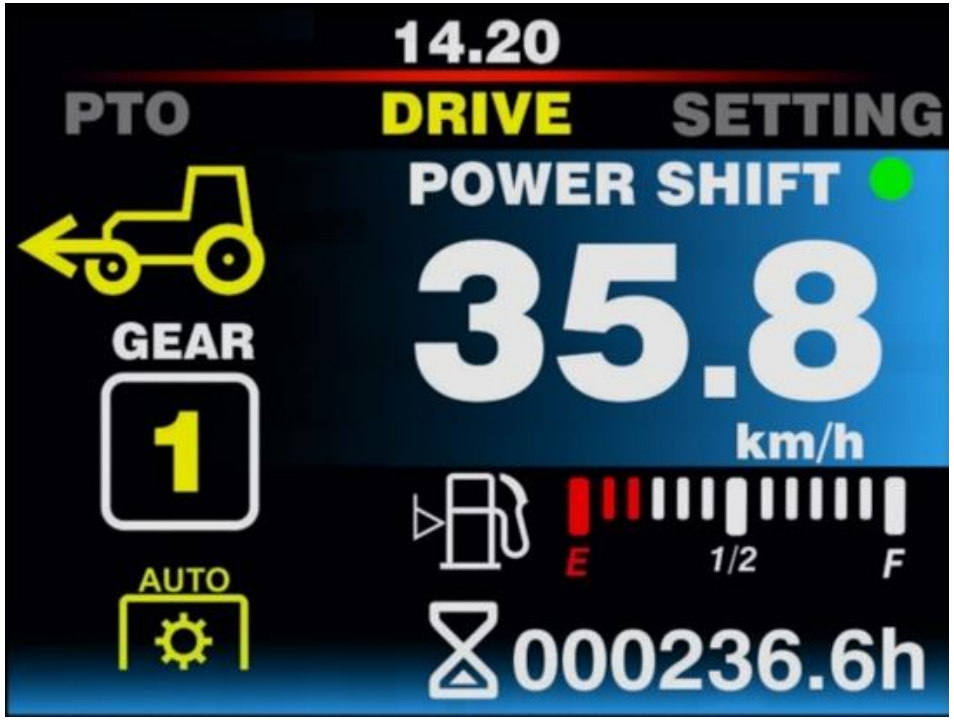
OUTPUT

- 1 x 500 mA low side drivers with analog diagnostic

DISPLAY DETAILS

TFT

- 3,5 ’’ color TFT LCD with optical bonding
- Normally black
- Active area: 70,08 x 52,56 mm
- 320 x RGB x 240
- 16.7M colors
- Pixel pitch: 0,219 x 0,219 mm
- Brightness: 800 cd/m^2 typ.
- Contrast ratio: 600 Min, 800 typ.
- Viewing angle: 80/80/80/80°
- Antiglare and antireflection treatments



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OPTIONAL VERSION

An optional version is available with two rubber buttons integrated in the front mask, without backlight.



2 rubber buttons

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DEVELOPMENT WITH MTASTUDIO

MTA Studio is an in-house developed code composing tool. It has been designed to program MTA's off the shelf line of dashboards, displays and ECUs. MTA Studio is powerful and easy to use. It empowers users to tailor device functionalities according to their needs.

MTA provides code examples that OEM engineers can edit and compile. The resulting code can then be uploaded to the cluster for final verification.

MTA can provide examples of MTA Studio projects from which OEM technicians can include specific graphics; the resulting code can then be uploaded to the cluster for final verification.





DEVELOPMENT WITH

MTA Studio is a tool used to program application software for dashboards, displays and Electronic Control Units.
It can be used on all MTA devices, custom and off-the-shelf.

APPLICATION LOGIC

- Logic functions programmed using graphical programming languages compliant to standard IEC 1131-3 (FBD – Function Block Diagram) or C/C++
- HMI designed using a WYSIWYG (What You See Is What You Get) interface. Segment, dot matrix and TFT display technology are supported
- Easy to use: autocomplete, help, find-variables and functions; clean graphic interface, simple application software interfaces: hardware is managed mainly through variables that are read and written
- Integrated debugger based over an XCP protocol
- Supported CAN bus interfaces: J1939 transport protocols (BAM/CMDT), programmable CAN wake up and termination
- Rich set of libraries: IEC 61131-3, odometer, hourmeter, etc.
- Real-time task management
- Code protection using HASP USB dongle to prevent unauthorized access
- Easy integration of external code and libraries

PROGRAMMED / CONFIGURED WITH

MTA Gate

MTA Gate is a tool designed for technical service personnel. Specifically designed to upload software upgrades and application software, it is also used to edit device configurations (EEPROM).

APPLICATION LOGIC

- Software programming / update: MTA Gate allows to transfer on all devices a Software generated by MTA Studio implementing controls on device “Hardware Variant ID*” and “Customer ID” to prevent misuse / wrong software updates
- Parameters setting: possibility to read and write EEPROM parameters on devices
- Access to device services: to take advantage of software application utilities such as stepper motor calibration
- Import from MTA Studio of parameters map
- MTA Gate can be used by standard Windows GUI or Command-Line for EOL process automation purposes
- MTA Gate license distribution through USB dongle

SPOT vs SPOT PLUS

FUNCTION	SPOT	SPOT PLUS
Display	3.1’’black & white dot matrix LCD: • 160 x 56 pixels • Brightness: 80 cd/m^2 max	3.5’’ color TFT with optical bonding: • 320 x RGB x 240 • Brightness: 800 cd/m^2 typ. • Contrast ratio: 600 min, 800 typ. • Viewing angle: 80/80/80/80°
Transparent cover	PMMA plastic with antifog treatment	Mineral glass with antiglare and antireflection treatments
Microcontroller	ST SPC560B54 with 64 MHz 32-bit PowerPC Flash: 768 KB; RAM: 64 KB	Traveo II CYT3DL with 240 MHz 32-bit Arm Cortex M7 CPU and 100 MHz 32-bit Arm Cortex –M0+ CPU Flash: 4,16 MB; RAM: 512 KB; Video RAM: 2 MB
Esternal Flash	Not available	32 MB
RTC (Real Time Clock)	Available on ‘full version’ without RTC retention in case of battery disconnection	Available with RTC retention in case of battery disconnection. Estimated time: 10 days @ 25 °C typical

SPOT vs SPOT PLUS

FUNCTION	SPOT	SPOT PLUS
Power supply	12 Vdc	12/24 Vdc
Dashboard thickness	1 – 15 mm	1 – 7 mm
Telltales	18 white LEDs	5 white LEDs (other telltales can be displayed on the TFT)
Internal buzzer	90 dB @ 30 cm	80 dB @ 30 cm

TIMELINE

- B-Sample: HW R1, TFT A-sample, Soft tools mechanics + MTA Studio beta release → **March 2026**
- End of Product Validation → **January 2027**
- SOP → **February 2027**

PRODUCT QUALIFICATION

The cluster will be qualified following the MTA TS50_16_rev_6 Test Specification which includes:

- Environmental tests
- Mechanical tests
- Chemical tests
- Electrical tests
- EMC tests

UNECE R10 Type approval included

THANK YOU

for your attention

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